SIGMACAMP · ELECTROCHEMISTRY LAB · SESSION 3

Silver Plating + Faraday's Law

Lecture & Instructor Notes

Audience: instructors and teaching assistants

Purpose: Teaching script, deep-dive notes, and coaching guidance

Session hook: Estimate how many atoms were deposited

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Lecture talking points
3. Instructor deep-dive notes
4. Preparation reminders

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

SESSION OVERVIEW & PLAN

Session 3 — Silver Plating and Faraday's Law

Hook: Calculate how many silver atoms you deposit and how thick the film is.

Learning outcomes

- Plate silver onto a metal object with controlled current and time
- Use Q = It, n = Q/F, and mass/thickness relations
- Discuss assumptions (100% efficiency, uniform coverage)
- Compare ideal Faraday prediction to visual result

Session plan (90 min)

Time	Activity	Reference
0–10 min	Hook: dull copper object — can we count the atoms we add?	lecture.md
10–25 min	Ag^+ reduction and Faraday's law	lecture.md
25–35 min	Estimate plated surface area	experiment.md — Area
35–45 min	Set up cell with ammeter in series	experiment.md — Setup
45–65 min	Plate with timed current readings	experiment.md — Run
65–80 min	Calculate charge, moles, mass, thickness	experiment.md — Calculations
80–90 min	Discuss efficiency and side reactions	lecture.md

Key equations

```
Ag⁺(aq) + e⁻ → Ag(s)
Q = I × t
n_e = Q / F     (F = 96,485 C/mol)
n_Ag = n_e
m_Ag = n_Ag × M_Ag   (M_Ag = 107.87 g/mol)
d = m_Ag / (ρ_Ag × A)   (ρ_Ag = 10.49 g/cm³)
```

Status

- [ ] $AgNO_3$ solution prepared (0.01 M)
- [ ] Calculation worksheet ready
- [ ] Pilot plating run completed
- [ ] Waste container labeled for silver

LECTURE TALKING POINTS

Session 3 — Lecture: Faraday's Law and Silver Plating

Target duration: ~20 minutes

> **Instructors:** worked Faraday example, z-factor for Ag vs Cu, efficiency — instructor.md.

Opening hook (0–10 min)

Show a dull copper coin or washer.

"If we plate this with silver for exactly 5 minutes at 0.02 A, can we calculate how many silver atoms landed on the surface — without weighing it?"

Answer preview: **Yes — with Faraday's law.**

Core concepts (10–25 min)

1. Silver reduction

```
Ag⁺(aq) + e⁻ → Ag(s)
```

- Each silver ion needs **one electron** to become silver metal
- So: **moles of electrons = moles of silver deposited** (1:1)

2. Charge, current, and time

Symbol	Meaning	Units
I	Current	Amperes (A) = C/s
t	Time	Seconds (s)
Q	Charge	Coulombs (C)

Q = I × t

Example: 0.020 A for 300 s → Q = 6.0 C

3. Faraday constant

F = 96,485 C/mol — charge of one mole of electrons

n_e = Q / F

Example: 6.0 / 96,485 = 6.22×10^{-5} mol e^-

4. From moles to mass to thickness

1. n_Ag = n_e (1 electron per Ag^+)

2. $m\_Ag = n\_Ag \times 107.87$ g/mol
3. Volume $V = m\_Ag / \rho\_Ag$ ($\rho\_Ag = 10.49$ g/cm^3)
4. Thickness $d = V / A$ (A = plated area in cm^2)

5. Worked example (on board)

Step	Calculation	Result
Q	0.020 A × 300 s	6.0 C
n_e	6.0 / 96,485	6.22×10^{-5} mol
m_Ag	n × 107.87	0.00671 g
V	m / 10.49	6.40×10^{-4} cm^3
d (A = 10 cm^2)	V / A	6.4×10^{-5} cm ≈ **0.64 μm**

6. Ideal vs. real plating

Faraday gives the **maximum** if every electron plates silver.

Reality may differ because:

- Side reactions consume current (H_2 evolution, etc.)
- Deposit may be porous or non-uniform
- Current may drift over time
- Not all surface area plates equally

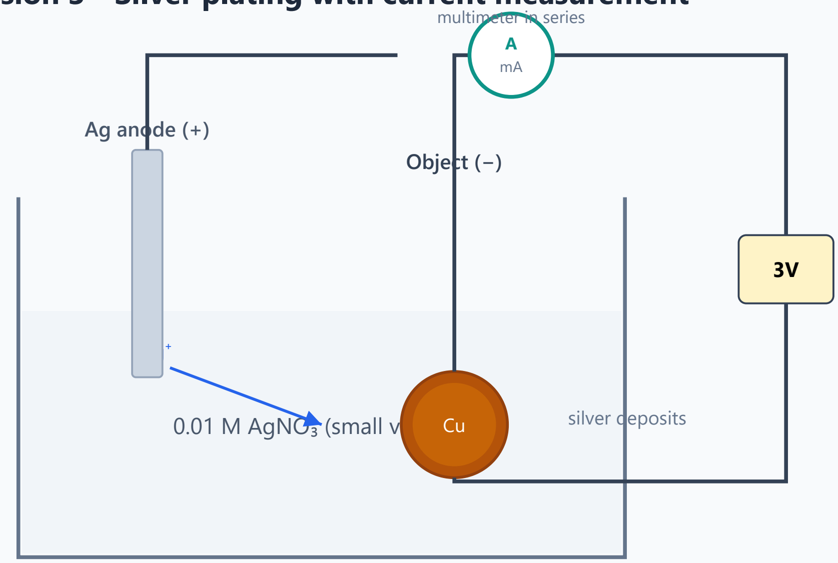
Wrap-up (80–90 min)

- How close was your calculated thickness to what you see?
- What would happen with a **silver anode** vs. inert anode?
- Where else is Faraday's law used? (electrolysis industry, battery design, analytical chemistry)

Visual aids

- [ ] Flowchart: I, t → Q → n → m → d
- [x] Plating setup (see experiment page and figure below)

Session 3 - Silver plating with current measurement



$Ag^+ + e^- \rightarrow Ag(s)$ Record I every 30-60 s and total time t

$Q = I \times t \rightarrow n = Q/F \rightarrow m\_Ag \rightarrow$ thickness $d = m/(\rho A)$

Estimate plated area A before starting (coin face, washer ring, etc.)

- [ ] Optional: stacked-atom illustration for "thickness"

Vocabulary

- [ ] Faraday constant
- [ ] Coulomb
- [ ] Stoichiometry of electron transfer
- [ ] Current efficiency (introductory)

Bridge to Session 4

"We've moved metal atoms with electricity. Next we split water molecules into hydrogen and oxygen — and measure the gases to prove the 2:1 ratio."

INSTRUCTOR DEEP-DIVE NOTES

Session 3 — Instructor Notes: Silver Plating + Faraday Deep Dive

Audience: instructors and TAs only. Pair with lecture.md and experiment.md.

Why this session exists

Students saw copper move in Session 2. Now they **quantify** electrodeposition: charge → moles of electrons → moles of metal → mass → thickness. Silver is ideal because $Ag^+ + e^- \rightarrow Ag$ is 1:1 (simpler stoichiometry than $Cu^{2+} + 2e^-$). This is the week's heaviest math session — go slowly on units.

Atomic / electronic foundations for silver

Silver ion and electron count

```
Ag⁺(aq) + e⁻ → Ag(s)
```

- Silver atom (Ag, Z = 47) loses one valence electron → Ag^+.
- Plating reverses that: one electron returns → one Ag atom deposited.
- Therefore: **n_Ag = n_e** (moles silver = moles electrons) when plating is 100% efficient.

Compare with copper from Session 2:

```
Cu²⁺ + 2e⁻ → Cu    ⇒   n_Cu = n_e / 2
```

Mention this contrast so students see why the electron count in the half-reaction matters.

Why dilute $AgNO_3$

Silver nitrate provides Ag^+. Classroom concentration **0.01 M** keeps:

- Cost down
- Staining / hazard manageable
- Thin films that still look metallic

Safety (non-negotiable): $AgNO_3$ stains skin/clothes permanently dark; gloves + goggles; collect all silver waste.

Faraday's law — instructor mastery

Charge from current and time

```
Q = I × t
```

Symbol	Unit	Meaning
I	ampere (A) = C/s	Current
t	second (s)	Time
Q	coulomb (C)	Total charge passed

Trap: students record mA but forget to convert. 20 mA = 0.020 A.

Faraday constant

```
F = 96,485 C/mol e⁻ ≈ 9.65 × 10⁴ C/mol
```

Meaning: one mole of electrons carries ~96,485 C.

```
n_e = Q / F
```

Stoichiometry factor (general form)

For a reduction requiring **z** electrons per metal ion:

```
n_metal = n_e / z = Q / (z F)
```

Metal ion	z	Example
Ag^{+}	1	Session 3
Cu^{2+}	2	Session 2
Al^{3+}	3	not used here

Mass and thickness

```
m = n × M
V = m / ρ
d = V / A
```

Quantity	Silver value
Molar mass M	107.87 g/mol
Density ρ	10.49 g/cm^3
A	geometric plated area (cm^2) — **biggest student error source**

Worked example (memorize this board version)

Given: I = 0.020 A, t = 300 s, A = 10 cm^2, Ag plating, z = 1.

Step	Math	Result
Q	0.020 × 300	6.0 C
n_e	6.0 / 96485	6.22×10^{-5} mol
n_Ag	= n_e	6.22×10^{-5} mol
m	6.22e−5 × 107.87	6.71×10^{-3} g
V	m / 10.49	6.40×10^{-4} cm^3
d	V / 10	6.4×10^{-5} cm = **0.64 µm**

Sense-check: visible silver films are often tenths of a micrometer to a few micrometers — students' numbers should be in that ballpark if I, t, A are sane.

Average current when I drifts

If students log I every 30–60 s:

```
I_avg ≈ (I₀ + I₁ + … + Iₙ) / (n+1)
```

or trapezoidal estimate. Then Q ≈ I_avg × t_total.

Do **not** use only the first reading if current falls (common as concentration polarization / surface changes).

Ideal vs real plating (current efficiency)

Faraday assumes **every** electron reduces Ag^+. Reality:

```
current efficiency η = (charge used for Ag plating) / (total charge) ≤ 1
```

Loss channels:

- H_2 evolution
- Impurity reductions
- Non-uniform current (edges plate more — "dog-boning")
- Flaky deposit that rinses off

If calculated thickness looks "too thick" vs visual: area underestimate or optimistic η = 1. If "too thin": efficiency < 1, area overestimate, or current logged wrong.

Electrochemical series note for Ag

Couple	E° (V)
Ag^+/Ag	+0.80
Cu^{2+}/Cu	+0.34

Couple	E° (V)
H^+/H_2	0.00

Ag^+ is a strong oxidant relative to H^+ and Cu^{2+} — silver plates readily. That is why even dilute $AgNO_3$ works visually. It is also why silver ions are “eager” to reduce and why waste must be handled carefully (and why Ag stains via reduction on skin/organics).

Anode choice:

- **Silver anode:** replenishes Ag^+ (like Cu anode in Session 2).
- **Inert anode:** Ag^+ depletes; oxygen evolution possible; calculations still OK for short runs if solution starts with enough Ag^+.

Experiment coaching

Area estimation — coach aggressively

Object	Reasonable approach
Coin face	$A = \pi(r)^2$ per face; ×2 if both faces plate
Washer	$\pi(R^2 - r^2)$
Irregular	Length × width of immersed region; say so

Students who invent a huge area get absurdly small thickness and think Faraday “failed.”

Ammeter in series

Multimeter must be **in series** on DC current range — not across the cell like a voltmeter (that shorts / misreads). Walk each station once.

Timing discipline

Start stopwatch when current is flowing and stable. Stop when disconnected. Gaps in logging → poor I_{avg}.

What success looks like

- Whitish / silvery sheen on copper object
- Not necessarily mirror-bright (classroom baths lack brighteners)
- Calculation completed with units at every step

Misconceptions

Misconception	Correction
“$Q = I + t$”	$Q = I \times t$
“Use mA directly in $Q = It$”	Convert to amperes

Misconception	Correction
“Cu and Ag use the same z”	Ag^+ needs $1e^-$; Cu^{2+} needs $2e^-$
“Thickness must match a ruler”	Micrometers are invisible to a ruler; appearance is qualitative
“Faraday is wrong if deposit looks thin”	Efficiency and area errors dominate

Board plan

1. Write $Ag^+ + e^- \rightarrow Ag$; circle “1 electron per atom.”
2. Flowchart: I,t → Q → n_e → n_Ag → m → V → d.
3. Full numerical example with units.
4. Discuss why I_avg beats a single reading.
5. Bridge to Session 4: next we count **gas molecules** by volume ratio, not metal atoms by charge.

Optional enrichment

- Derive z from half-reaction for Cu plating retroactively (Session 2 data).
- Mention electrolysis mass production (Al from Hall–Héroult — huge Faraday consumer).
- Introduce faradaic efficiency formally if a student is ready.

PREPARATION REMINDERS

Session 3 — Preparation

Before class (instructor)

- [ ] Prepare 0.01 M $AgNO_3$ with **distilled/deionized water** (not tap); volume for all groups + 20% extra
- [ ] Pilot plate one object: record I_avg, t, appearance
- [ ] Verify 3 V packs deliver stable ~10–30 mA in this cell
- [ ] Print calculation worksheets
- [ ] Confirm silver waste disposal with school/facility protocol

$AgNO_3$ handling

- Wear gloves when preparing
- Work in ventilated area
- Label all bottles clearly
- Store away from organic materials and light if possible

Day-of setup

- [ ] Small volumes pre-measured per station (reduces spills)
- [ ] Goggles enforced at door
- [ ] Waste container visible and labeled
- [ ] Demo calculation on board with example numbers (before students start)

Timing note

Calculations take longer than expected for some groups. Have I_avg pre-computed on board as check, or provide a spreadsheet template.

Open questions / notes

- Stock $AgNO_3$ concentration on hand:
- Volume prepared (mL):
- Pilot I_avg (mA):
- Pilot plating time (s):